

Thermodynamics Formula Sheet

Final Exam

Units and Conversions

SI Prefixes

femto (f)	= 10^{-15}	deca (da)	= 10^1
pico (p)	= 10^{-12}	hecto (h)	= 10^2
nano (n)	= 10^{-9}	kilo (k)	= 10^3
micro (μ)	= 10^{-6}	mega (M)	= 10^6
milli (m)	= 10^{-3}	giga (G)	= 10^9
centi (c)	= 10^{-2}	tera (T)	= 10^{12}
deci (d)	= 10^{-1}	peta (P)	= 10^{15}

Atmospheric Pressure

$$\begin{aligned}
 1 \text{ atm} &= 101.325 \text{ kPa} &= 29.92 \text{ inHg} \\
 &= 1.01325 \text{ bar} &= 14.696 \text{ psi} \\
 &= 760 \text{ Torr} &= 10.33 \text{ m H}_2\text{O (at 4}^\circ\text{C)} \\
 &= 760 \text{ mmHg}
 \end{aligned}$$

Pressure Conversions

$$\begin{aligned}
 1 \text{ Pa} &= 1 \text{ N/m}^2 \\
 1 \text{ bar} &= 10^5 \text{ Pa} \\
 1 \text{ psi} &= 6.89476 \times 10^3 \text{ Pa} \\
 1 \text{ Torr} &= 133.322 \text{ Pa} \\
 1 \text{ mmHg} &= 133.322 \text{ Pa} \\
 1 \text{ inHg} &= 3.38639 \times 10^3 \text{ Pa}
 \end{aligned}$$

Gauge vs Absolute Pressure

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmospheric}}$$

$$P_{\text{vacuum}} = P_{\text{atmospheric}} - P_{\text{absolute}}$$

Note: Generally, gauge pressures already account for atmospheric pressure, and thus reads zero when open to the atmosphere.

Pressure Relations

$$P = \frac{F}{A} = \rho gh \quad (\text{hydrostatic pressure})$$

$$P_1 = P_2 \quad \Rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad (\text{Pascal's Law})$$

$$\Delta P = \rho gh \quad (\text{multi-fluid manometer})$$

(add downward, subtract upward)

Density and Specific Gravity

$$\rho = \frac{m}{V} \quad (\text{density}), \quad \vartheta = \frac{1}{\rho} = \frac{V}{m} \quad (\text{specific volume})$$

Specific gravity is defined as a relative density compared to water (at 4°C) where $\rho_{\text{water}} = 1000 \text{ kg/m}^3$:

$$SG = \frac{\rho_{\text{fluid}}}{\rho_{\text{water}}} \Rightarrow \rho_{\text{fluid}} = SG \cdot \rho_{\text{water}}$$

Specific Weight

$$\gamma_s = \rho g \quad (\text{specific weight})$$

Where hydrostatic pressure can be written as $P = \gamma_s h$

Multi-fluid Manometer

1. Begin at a known pressure point (gauge or atmospheric pressure) and follow the fluid layers to the unknown pressure point.
2. Sign convention: add when going down, subtract when going up.
3. **Horizontal jump:** We can "jump" horizontally across bends in the tube if both sides of the jump are within the **same continuous fluid**. This is because pressure is identical at the same horizontal level within a single static fluid. *Any pressure decrease from moving upward is perfectly balanced by an equal pressure increase when moving back down to that same level on the other side.*
4. The pressure at each end of the manometer equals the total pressure obtained by summing the contributions of all fluid columns along the vertical path.

Pascal's Principle

The pressure applied to a confined fluid increases the pressure throughout the fluid by the same amount.

Temperature Conversions

- Absolute temperature conversions
 - $T_K = T_{^\circ C} + 273.15$
 - $T_{^\circ R} = T_{^\circ F} + 459.67$
 - $T_{^\circ F} = \frac{9}{5}T_{^\circ C} + 32$
 - $T_{^\circ R} = \frac{9}{5}T_K$
- Temperature difference conversions
 - $\Delta K = \Delta ^\circ C$
 - $\Delta ^\circ R = \Delta ^\circ F$
 - $\Delta ^\circ F = \frac{9}{5}\Delta ^\circ C$
 - $\Delta ^\circ R = \frac{9}{5}\Delta K$

Volume

$$1 \text{ m}^3 = \underline{1000 \text{ L}} = 61024 \text{ in}^3 = 35.315 \text{ ft}^3 = \underline{264.17 \text{ gal}}$$

Slugs and lbm

$$\left[\frac{1 \text{ lbf}}{32.174 \frac{\text{lbm} \cdot \text{ft}}{\text{s}^2}} \right], \quad \left[\frac{1 \text{ slug}}{32.174 \text{ lb}} \right], \quad \left[\frac{1 \text{ lbf}}{1 \frac{\text{slug} \cdot \text{ft}}{\text{s}^2}} \right]$$

Energy Units

$$1 \text{ Btu} = 778.169 \text{ ft} \cdot \text{lbf} = 1055.06 \text{ J}$$

Work Units

$$1 \text{ hp} = 550 \text{ ft} \cdot \text{lbf/s} = 745.7 \text{ W}$$

Distance

$$1 \text{ mile} = 5280 \text{ ft}$$

Ideal Gas Equation of State

Refer to table A-4 for gas constants of common gases.

$$P\bar{v} = RT, \quad R = \frac{\bar{R}}{M} \quad \left(\text{kJ/kg} \cdot \text{K} \quad \text{or} \quad \frac{\text{Btu/lbm} \cdot \text{R}}{\text{psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R}} \right)$$

$$P\bar{V} = mRT \quad (\text{in terms of TOTAL volume and mass})$$

Universal gas constant: $\bar{R} = 8.314 \text{ J}/(\text{mol} \cdot \text{K})$

Note: Convert units for P , $\bar{V}$, $\bar{v}$, and T to match the units of R when using the ideal gas law. Also, $[\text{kJ/kPa}] = [\text{m}^3]$

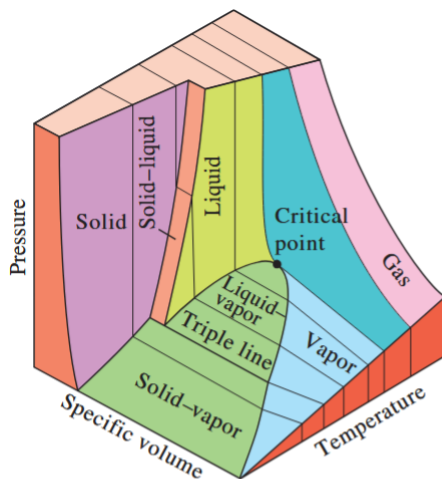
Work & Heat Sign Convention

	IN	OUT
Heat Q	+	-
Work W	-	+

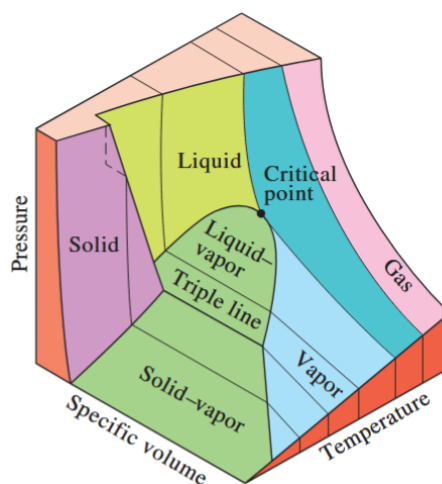
Linear Interpolation Template

Interpolation using values at (T_1, y_1) and (T_2, y_2) :

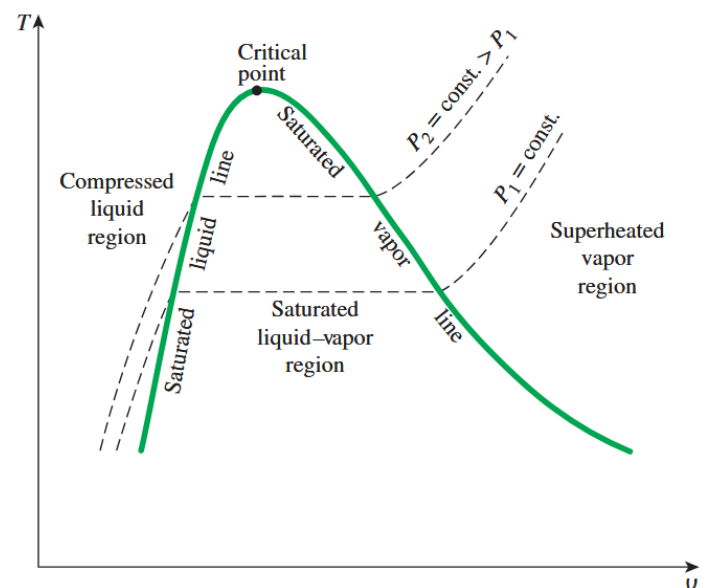
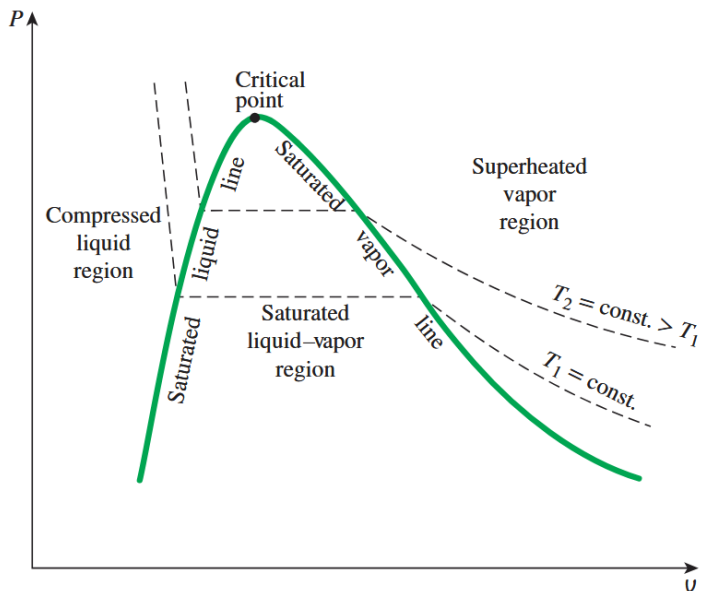
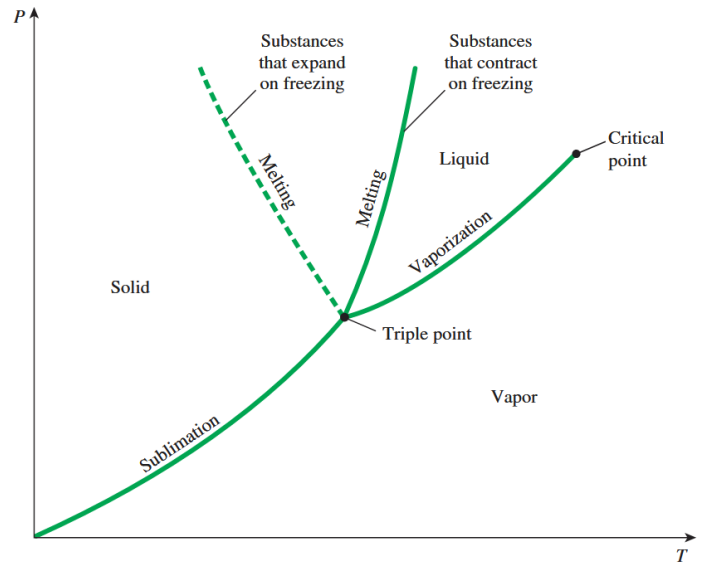
$$y - y_1 = \frac{y_2 - y_1}{T_2 - T_1}(T - T_1)$$



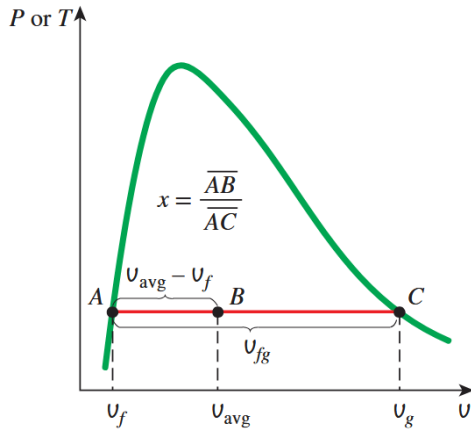
P - v - T surface of a substance that contracts on freezing.



P - v - T surface of a substance that expands on freezing (like water).



Phase Change & Saturation



$$h_{fg} = h_g - h_f \quad (\text{latent heat of vaporization})$$

$$x = \frac{m_g}{m_f + m_g} \quad (\text{quality definition})$$

$$\left. \begin{aligned} y_{avg} &= y_f + x(y_g - y_f) \quad (\text{for } \phi, h, u) \\ x &= \frac{y_{avg} - y_f}{y_g - y_f} \quad (\text{solve for quality}) \end{aligned} \right\} y_{fg} = y_g - y_f$$

Compressed Liquid Approximation

$$y(T, P) \approx y_f(T) \quad (\text{use saturated liquid at same } T)$$

Enthalpy

$$h = u + P\phi, \quad \phi = \frac{1}{\rho} \quad (\text{specific enthalpy})$$

Specific Heat Capacity

$$c_\phi = \left(\frac{\partial u}{\partial T} \right)_\phi \quad (\text{fixed volume})$$

$$c_p = \left(\frac{\partial h}{\partial T} \right)_p \quad (\text{fixed pressure})$$

$$\Rightarrow \Delta u = c_\phi \Delta T, \quad \Delta h = c_p \Delta T \quad (\text{constant } c_\phi \text{ or } c_p)$$

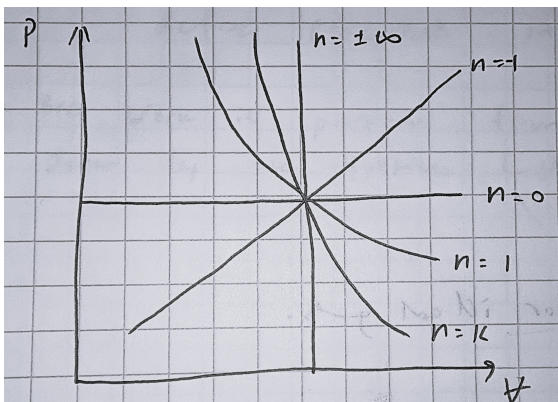
$$c_p = c_\phi + R \quad (\text{ideal gas})$$

$$c_p = c_\phi \quad (\text{incompressible substances})$$

$$Q = mc\Delta T \quad (c_\phi \text{ or } c_p \text{ depending on the process})$$

$$k = c_p / c_\phi \quad (\text{specific heat ratio}), \quad c_{\text{water}} = 4.18 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

Polytropic Processes



$$Pv^n = \text{constant}$$

- $n = 0$: isobaric process (constant pressure)
- $n = 1$: isothermal process (constant temperature, ideal gas)
- $n = k$: isentropic process (constant entropy)
- $n = \infty$: isochoric process (constant volume)

First Law (Closed System)

$$\Delta U = Q - W \quad (\text{energy balance for closed system})$$

$$\dot{E}_{in} - \dot{E}_{out} = \frac{dE_{system}}{dt} \quad (\text{rate form})$$

First Law (Open System)

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \underbrace{\sum \dot{m}_i \left[h_i + \frac{1}{2} V_i^2 + gz_i \right]}_{\text{Inlet}} - \underbrace{\sum \dot{m}_e \left[h_e + \frac{1}{2} V_e^2 + gz_e \right]}_{\text{Outlet}}$$

Note:

$$\dot{W} = \dot{m} \frac{\Delta P}{\rho} = \dot{V} \Delta P \quad (\text{pump power w/ no change in KE or PE})$$

Mass & Flow Relations

$$\dot{m} = \rho \dot{V} = \rho A v_{avg} \quad (\text{mass flow rate})$$

$$\dot{E} = \dot{m} e \quad (\text{energy flow rate})$$

Work

$$W = \int F \cdot dx, \quad \dot{W} = F \cdot v \quad (\text{general})$$

$$\begin{aligned} W &= \int P dV \Rightarrow W = \frac{P_2 V_2 - P_1 V_1}{-n + 1} \quad (\text{polytropic, } n \neq 1) \\ &\Rightarrow W = C \ln \left(\frac{V_2}{V_1} \right) \quad (\text{isothermal, } n = 1) \\ &\Rightarrow W = P \Delta V \quad (\text{isobaric}) \end{aligned}$$

Where C is $P_1 V_1^n = P_2 V_2^n = C$ for a polytropic process.

$$W = \frac{1}{2} k (x_2^2 - x_1^2) \quad (\text{spring work})$$

Heat Transfer

$$Q = \int \dot{Q} dt = \dot{Q} \Delta t \quad (\text{total heat transferred})$$

$$\dot{Q} = kA \frac{\Delta T}{\Delta x} \quad (\text{conduction})$$

$$\dot{Q} = hA(T_s - T_\infty) \quad (\text{convection})$$

$$\dot{Q} = \epsilon \sigma A (T_s^4 - T_\infty^4) \quad (\text{radiation})$$

Efficiency

Heat engine

$$\eta_{HE} = \frac{W_{HE}}{Q_H} = \frac{Q_H - Q_L}{Q_H} \leq \eta_{Carnot, HE} = 1 - \frac{T_L}{T_H}$$

Heat Pump

$$\beta_{HP} = \frac{Q_H}{W_{HP}} = \frac{Q_H}{Q_H - Q_L} \leq \beta_{Carnot, HP} = \frac{T_H}{T_H - T_L}$$

Refrigerator

$$\beta_{Ref} = \frac{Q_L}{W_{Ref}} = \frac{Q_L}{Q_H - Q_L} \leq \beta_{Carnot, Ref} = \frac{T_L}{T_H - T_L}$$

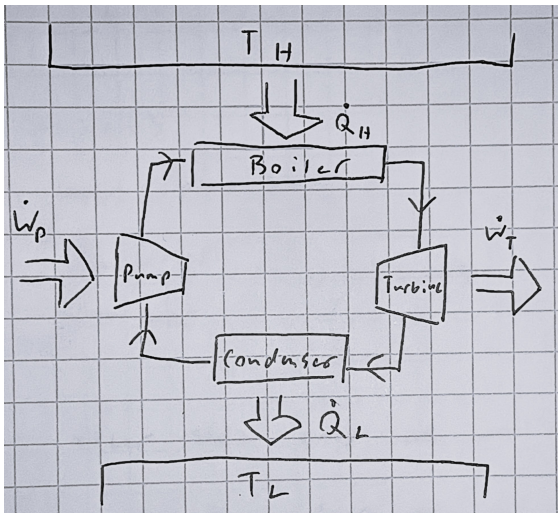
Note: The heat and work terms can also be in rate forms $\dot{Q}$ and $\dot{W}$.

T_H and T_L are ABSOLUTE temperatures!!

Heat Engines

Kelvin-Planck: there has to exist losses as heat to surroundings (Q_L).

Produces work from energy input as heat (Q_H).

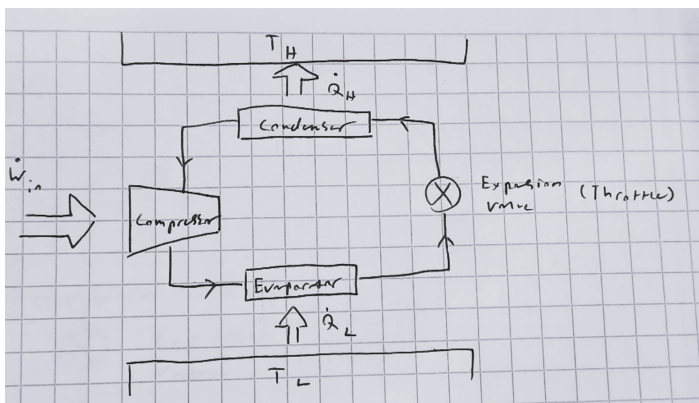


Note: $\dot{W}_{net} = \dot{W}_T - \dot{W}_P$

Refrigerators & Heat Pumps

Clausius: requires work input to transfer heat from cold to hot reservoir.

Refrigerators transfer heat from cold to hot reservoir, while heat pumps transfer heat from cold to hot reservoir. They have the same cycle but differ in *intent*.



Four Processes in a Carnot Cycle

For example, in a Carnot heat engine:

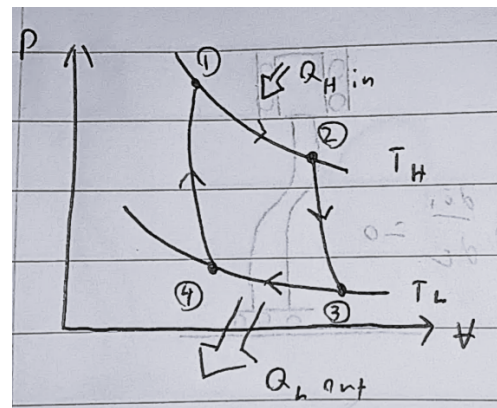
- 1 → 2: Reversible isothermal heat addition (Q_H in at T_H)
- 2 → 3: Reversible adiabatic expansion ($T \rightarrow T_L$)
- 3 → 4: Reversible isothermal heat rejection (Q_L out at T_L)
- 4 → 1: Reversible adiabatic compression ($T \rightarrow T_H$)

Carnot components are reversible and ideal, so they allow heat transfer even though they have no temperature difference.

All Carnot cycles operating between the same two temperature reservoirs have the same efficiency (determined by T_H and T_L only).

Ideal Gas Carnot Cycle

Consider the following Carnot cycle with an ideal gas in a piston-cylinder system:



Two isothermal processes ($dT = 0$): 1 → 2 and 3 → 4
Two adiabatic processes ($dQ = 0$): 2 → 3 and 4 → 1.

From ($P\varphi = RT$), ($\delta w = Pd\varphi$), and ($du = c_\varphi dT$):

$$\delta q = c_\varphi dT + \frac{RT}{\varphi} d\varphi \quad (\text{general ODE})$$

Integrating the ODE:

$${}_1q_2 = RT \ln \left(\frac{\varphi_2}{\varphi_1} \right) \quad (\text{for an isothermal process})$$

$$\ln \left(\frac{T_2}{T_1} \right) = -\frac{R}{c_\varphi} \ln \left(\frac{\varphi_2}{\varphi_1} \right) \quad (\text{for an adiabatic process})$$

Note:

For the adiabatic process, we assume a constant c_φ , but for large temp changes, use $c_{\varphi, avg} = (c_\varphi(T_1) + c_\varphi(T_2))/2$.

Clausius Inequality

For ALL cycles (reversible or irreversible):

$$\oint \frac{\delta Q}{T} \leq 0, \quad = 0 \text{ for reversible, } < 0 \text{ for irreversible}$$

$\oint \frac{\delta Q}{T} > 0$ is **impossible** since it violates the second law of thermodynamics

Entropy Definition

For a reversible process:

$$dS \equiv \left(\frac{\delta Q}{T} \right)_{\text{rev}} \left[\frac{\text{kJ}}{\text{K}} \right], \quad ds \equiv \left(\frac{\delta q}{T} \right)_{\text{rev}} \left[\frac{\text{kJ}}{\text{kg} \cdot \text{K}} \right]$$

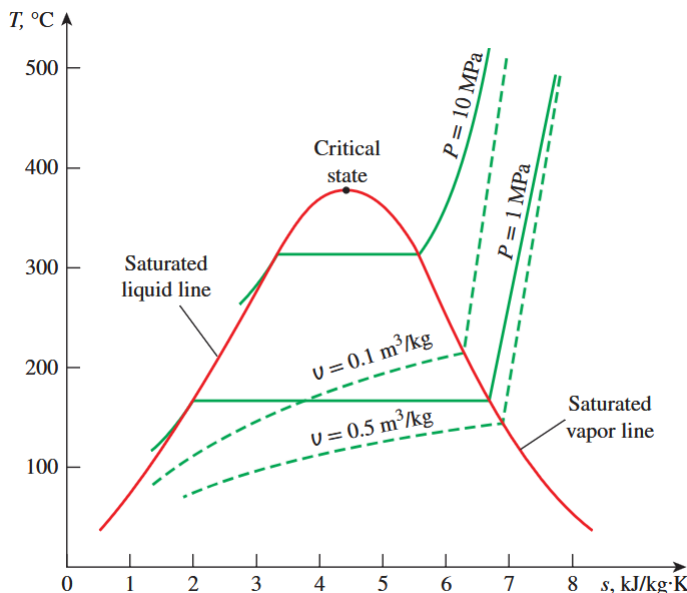
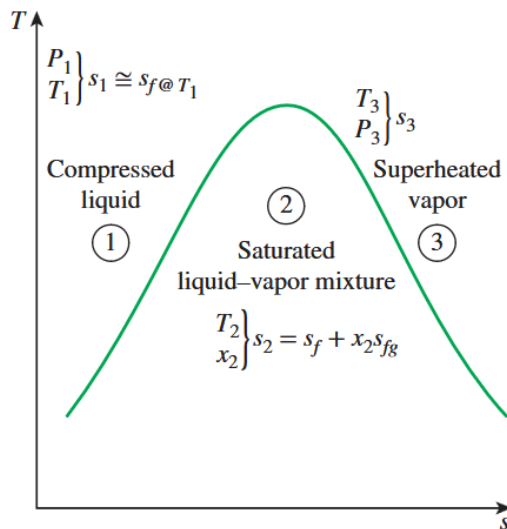
For a general process (reversible or irreversible):

$$S_2 - S_1 = \int_1^2 \left(\frac{\delta Q}{T} \right)_{\text{CS}} + {}_1S_2 \text{ gen}, \quad {}_1S_2 \text{ gen} \geq 0$$

$$\frac{dS_{\text{CM}}}{dt} = \frac{\dot{Q}}{T_{\text{CS}}} + \dot{S}_{\text{gen}}, \quad \dot{S}_{\text{gen}} \geq 0$$

Q refers to the heat transfer across the system boundary and T_{CS} is the temperature at the system boundary where the heat transfer occurs.

T-S Plots



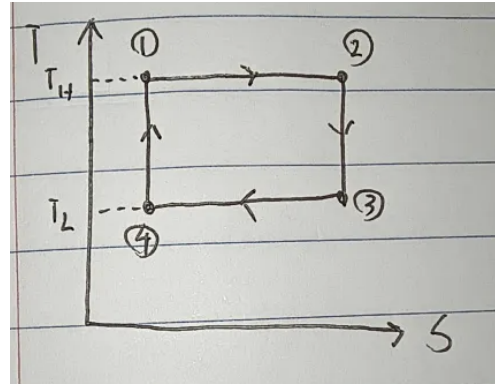
Heat Transfer for a Reversible Process

$$Q = \int T dS \quad (\text{area under the T-S plot})$$

Entropy Change for a Phase Change

$$s_{\text{fg}} = \frac{h_{\text{fg}}}{T_{\text{sat}}}$$

Carnot Cycle (Reversible) on T-S Plots



Isothermal processes are horizontal lines, and adiabatic processes are isentropic (vertical) lines (since $dS = \delta Q/T = 0$ for reversible adiabatic processes).

Gibbs Relations

General thermodynamic relationships between entropy and other properties:

$$T ds = du + P dv \quad (\text{First Gibbs Relation})$$

$$T ds = dh - v dP \quad (\text{Second Gibbs Relation})$$

Gibbs Relations for Ideal Gas

From First Gibbs Relation:

$$ds = \frac{C_{v0}}{T} dT + \frac{R}{v} dv$$

$$\Rightarrow s_2 - s_1 = C_{v0} \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$$

From Second Gibbs Relation:

$$ds = \frac{C_{p0}}{T} dT - \frac{R}{P} dP$$

$$\Rightarrow s_2 - s_1 = C_{p0} \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right)$$

Note that these two equations are equivalent since $C_{p0} = C_{v0} + R$ for an ideal gas.

Isentropic Relations for Ideal Gases ($Pv^k = \text{constant}$)

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}}, \quad \frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{k-1}, \quad \frac{P_2}{P_1} = \left(\frac{v_1}{v_2} \right)^k$$

Entropy Balance

For a CLOSED system:

$$\sum \frac{Q_k}{T_k} + S_{\text{gen}} = \Delta S_{\text{sys}} \quad [\text{kJ/K}]$$

For an OPEN system:

$$\sum \frac{\dot{Q}_k}{T_k} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{S}_{\text{gen}} = \frac{dS_{\text{CV}}}{dt} \quad [\text{kW/K}]$$